

IUS
INSTITUT
UNIVERSITAIRE
DES SCIENCES

Le rapport du laboratoire système

FACULTÉ DES SCIENCES ET TECHNOLOGIES

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NIVEAU : L3-FST

ENSEIGNANT : Ismaël SAINT_AMOUR

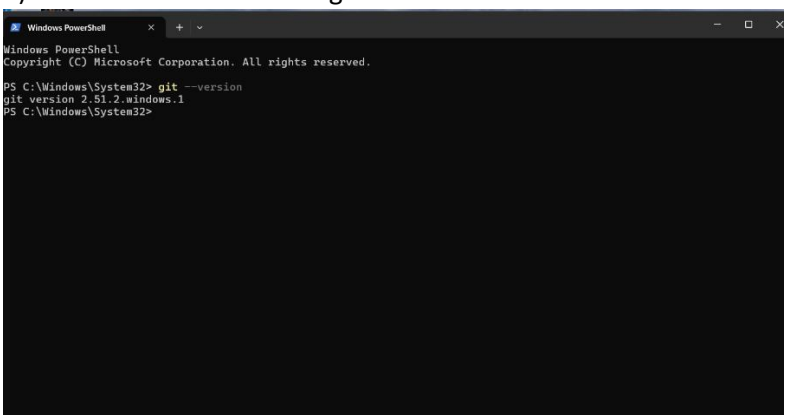
DISCIPLINE : Système

DATE : 19 octobre 2026

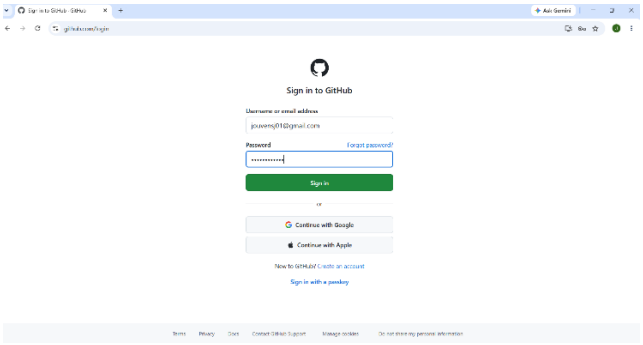
L’objectif de ce TD est de :

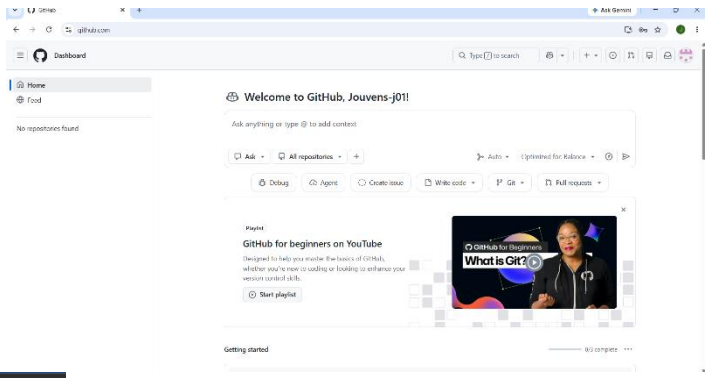
1) Comprendre les concepts de base de Git (système de gestion de versions).
2) Savoir utiliser pour gérer un projet localement.
3) Découvrir GitHub et apprendre à collaborer.
4) Reproduire une série de taches 1 à 8.
5) Créer un nouveau dépôt sur GitHub.

1)Vérification de la version git installe

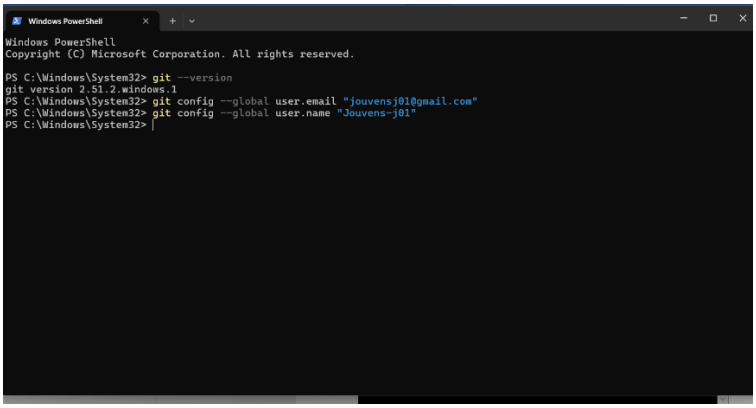


2) connexion avec mon compte GitHub et affichage de l'interface de mon compte

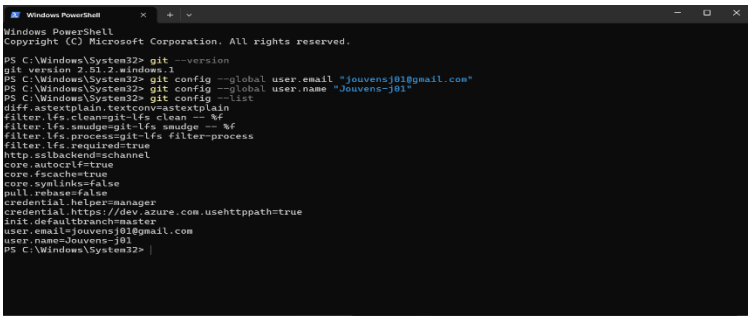




3) configuration de git

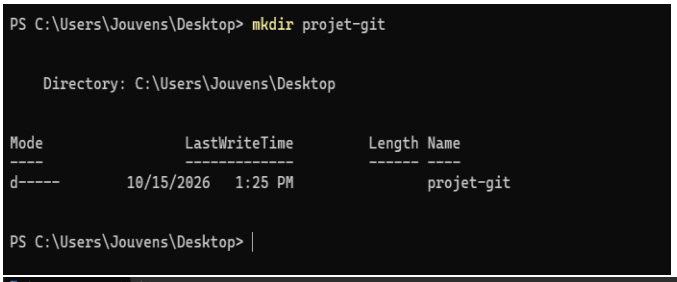


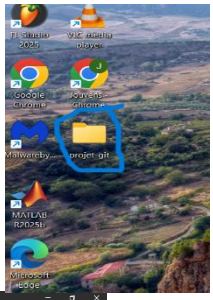
4) vérification des données utilisateurs

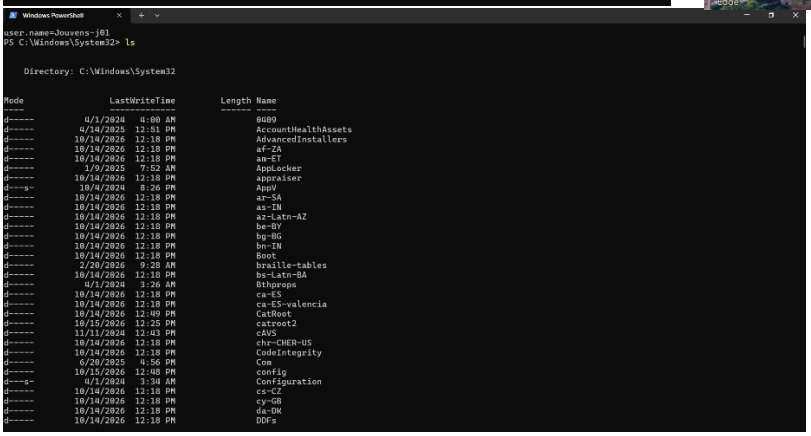


5) **Tester Git avec PowerShell**

a) Affichage de la liste et dossier avec la commande ls





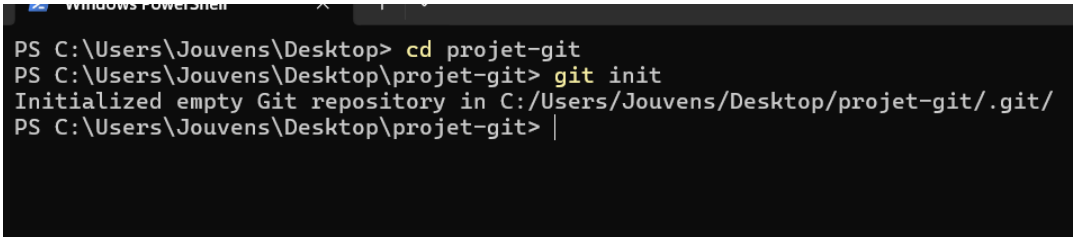


b) La création d'un fichier dans desktop

6) Entrée dans notre dossier «projet-git »



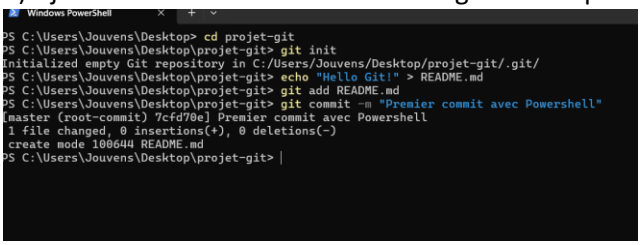
7) Initialisation du dépôt Git

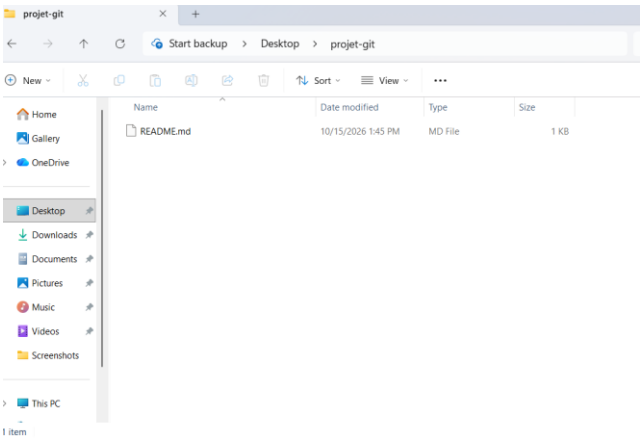


8) Ouverture et écriture de mon projet README.md

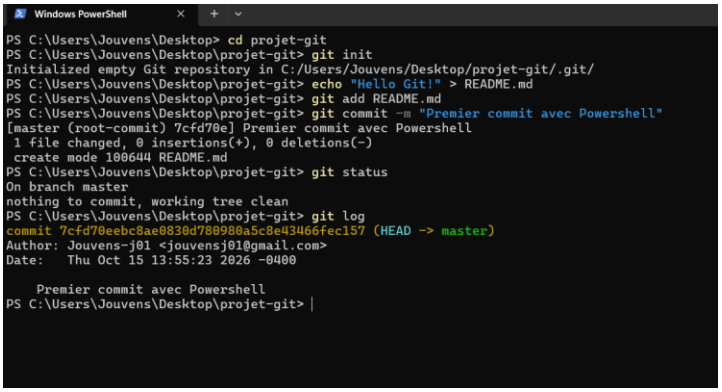


9) Ajout du dossier README.md dans git et mon premier commit effectuer avec PowerShell

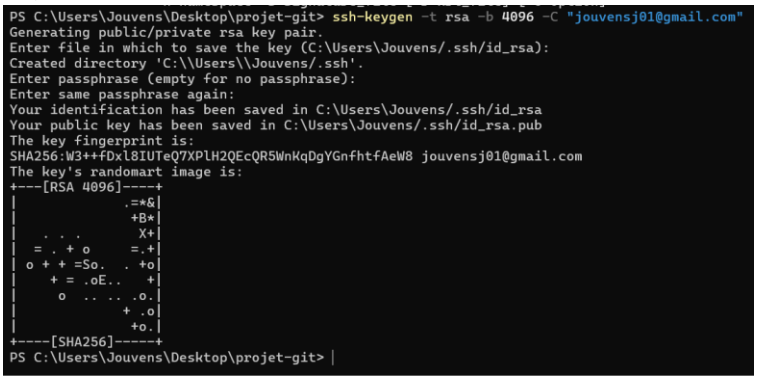




10) Affichages de l’historique des commits



11) Génération d’une nouvelle clé SSH



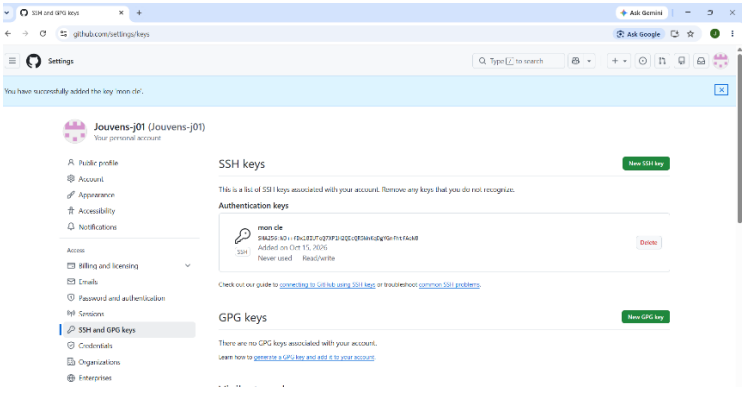
12) Affichage de la clé privée avec la commande cat

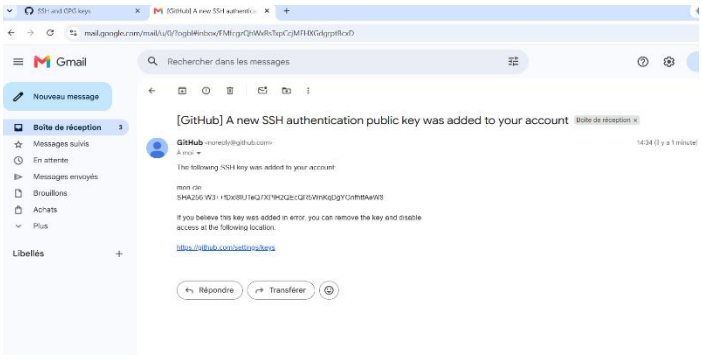
```
PS C:\Users\Jouvens\Desktop\projet-git> cat ~/.ssh/id_rsa
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAACFwAAAAdzc2gtcn
NhAAAAAwEAAQAAAgEA0dpkeHwvARJuCPlqQjTbuW7fKHTQDtAxqbO7xVw5iYQ+3TqjCE3a
d/6MWOsWvaIoKtsxatQIzZuRYd1uMB9iOmNsT/Dh52hKXOlFR4A5uAR4YKyEMRTeE65jbL
pMRxLK9w0U/FMiDcOUOEwqzrrPkXHWlXtLYQlFlJRtUpJ0CymV5vyy3MH9bHi5nAQAFt+u
7mnhWVYZYuhjIMHPfHiv8xygiRBf8c7lbsxQYfkLbTmi8PfsPg2pDFmmp6l8DrlmN4eqcS
EPgTw0bfTnJvfwYqEXQaeLSPPP8ToTY5Ep8roz1YNE5M1QH+O0csZ1gJP6Ts3tfNU3wexY
aadXK4FsiXtk13/KDy04a0HohYcqbtuBr5s2Jpyyybdt8Ht1kHriM2bAmNnrAZd8QmDxAZ
WLYcbgp8S62bM/NLDSGiOCZ8sAByWEmYE90dyIEl+WE+Z0SERGRUq2894BM2SP3wRCCqcb
M4jpbBpDq4CMePc1mrKuNdj9bzF4xPzQG5zBvWJYCBGNxZA/kBltL/DS9nqZajig/LaHkX
kXNghLuLuffb/Ii0mrmc1yGrH1tx9x80nBfZ9v+t+5Ev2Er7TizwHhUMwzv++1vpUQjs83
GJKj1Hqq96udOnfmPnowmkZckU1t1vtUOC7qmG6QzyuGJQKiRCTtqrF9Xzpz2tF3WpwIoK
cAAAdQ10z58NdM+fAAAAAHc3NoLXJzYQAAAgEA0dpkeHwvARJuCPlqQjTbuW7fKHTQDtAx
qbO7xVw5iYQ+3TqjCE3ad/6MWOsWvaIoKtsxatQIzZuRYd1uMB9iOmNsT/Dh52hKXOlFR4
A5uAR4YKyEMRTeE65jbLpMRxLK9w0U/FMiDcOUOEwqzrrPkXHWlXtLYQlFlJRtUpJ0CymV
5vyy3MH9bHi5nAQAFt+u7mnhWVYZYuhjIMHPfHiv8xygiRBf8c7lbsxQYfkLbTmi8PfsPg
2pDFmmp6l8DrlmN4eqcSEPgTw0bfTnJvfwYqEXQaeLSPPP8ToTY5Ep8roz1YNE5M1QH+O0
csZ1gJP6Ts3tfNU3wexYaadXK4FsiXtk13/KDy04a0HohYcqbtuBr5s2Jpyyybdt8Ht1kH
riM2bAmNnrAZd8QmDxAZWLYcbgp8S62bM/NLDSGiOCZ8sAByWEmYE90dyIEl+WE+Z0SERG
RUq2894BM2SP3wRCCqcbM4jpbBpDq4CMePc1mrKuNdj9bzF4xPzQG5zBvWJYCBGNxZA/kB
ltL/DS9nqZajig/LaHkXkXNghLuLuffb/Ii0mrmc1yGrH1tx9x80nBfZ9v+t+5Ev2Er7Ti
zwHhUMwzv++1vpUQjs83GJKj1Hqq96udOnfmPnowmkZckU1t1vtUOC7qmG6QzyuGJQKiRC
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JeAdv26aV+XQMx8NYWEPGbMQ9y+Z6Tlh1HAYmsjys7/DVgW1mP7TErYf2gqsLaBTiNdAgJ
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8cyyek7pKuyyAO+c5oCsbOaGd4RGfEQSu5YILg2V3AbbxZGnrTKg07UO0PuLVPEHVQ23ig
t9DtU98ki9mxixXQgNOzKeiUNvh27OJ74+0R5XQ3YBFacvAD3EO2lv6M1QYGGnyuvA3Dl6
5JXPATGKKramIpQBQrxaTmzqlZXJngT5YvGjYy/QTzI4h0aGJ0mEv6zK9BnDBgH/lIRBrJ
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n3YidUY3/UdAB/I6d6Qv6dyBzz/2JjkzDjpCT5flsa+3WqOaCC80GI/ctiDOHaM7W6Z11a
CAZ9hJzmjHR9A8A11ZRRYewRiUKJ3fNi0ijvy1yNuryqbbhHiMyw17qwUdrO+NZLHUH2YH
ThesSwYcwyhYtGvBYt1+qjqIA+t/WtY3pmgWWqVars5W8FUvB72uU1zZ09/VPyNysyRdnn
pQnS99UVgntZ2chstpAAABAF8dWVXfRM7q2fuvbxeLNwTk+u5yKeBZVdxEGQxxvOi4p40j
QXCk67GQomsu/r6qy3KJL9HQ3J1B7caYYnFnyLxKuhtaIdKE55HuHEwhZIHN0MwwT1wumX
LvaKmRPqBKmX0lXfcLtrQ4rxy1aWHMAnu+ELIZOH/ReStmQClq2jb6NqRguL1Wuvmn8lip
uaMoMgrL7uwvk+qdUJKu1Ik4UB8cBfdcb8rR+6XkoINNWGeTeXXVido61F9DpFqRydsgA/
wleJJ5LsOjoxhHI6lJw531kj8NCusJiq+pBQT3WCAzJZhhLihtTUdwBLkK65R42AY4MlrY
QpO7LXXjQppJBiIAAAEBAPO+LZDZK7RFvqFqI91YDCk1PPnQLhnc/VpaWJ9w7M/PiFE45X
XAZRAvVbvJSG3PNO8OLMt8rGf0SiN1ZSgbhi4J/msg6XF4MRL83VwkTkypyEFqxO9om6RL
IKbqjbKUPALPyRJCD97ywqICk0Llv8cnWSy++TWfULjM32l9n82/JiwVlscKii6SqXvFB
ZO4TEefKi1XE4eDdFj7mBSrZWL2zwi87yJW7U4ytKDVSY8PFK/cZDSGuJvsIQeWY8PC9Wi
9UERsg6dyzaRmEvg2NOa/WI6daNT4gX9zisvpAJ57y3bfhgL40YV+uLjEqPa4rgFTvHa3B
8JoxBxPfshIwsAAAEBANxn70AynM/DY0mnOddF3/lHtKZP2oGA97ytuCWS5WraMnLlW35B
sw6IIOeIMbeXxeRs+C9JS3OsdBgfHssRb71JbWc7b5oU98prXSvSyaZQ/yoJXUMPhb73yU
3/gnWO5hxsp3xUWWs2i52D8yyTl/OFqiMBKgq3lkZJQ554Yq9bCs0cSejgYEEiWqpRXYdE
pTiq8clHKjY8uqhkcDvO4hs8eAkY3V1kIHJ2kzcAAaD3nvJNbuRWHQcL/74uH/mJZJUd2B
A2+DDSycCri55VDMpr7tjx+OwV5X6YINg1yQ2RdM18NpMywhBs9aZ4+qS2O93YhmF30dO7
z+XxVa72ulUAAAAUam91dmVuc2owMUBnbWFpbC5jb20BAgMEBQYH
-----END OPENSSH PRIVATE KEY-----
PS C:\Users\Jouvens\Desktop\projet-git>
```

13) Affichage du clé publique

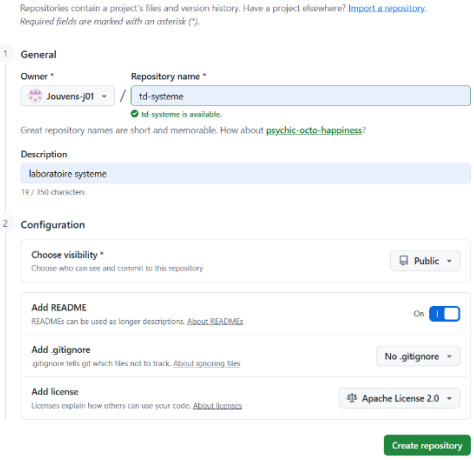
```
PS C:\Users\Jouvens\Desktop\projet-git> cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQDR2mR4fC8BEm4I+WpCNNu5bt8odNAO0DGps7vFXDmJhD7dOqMITdp3/oxY6xa9oigq2zFq1AjNm5Fh3W4wH2I6Y2xP8OHnaEpc6UVHgDm4BHhgrIQxFN4T
rmNsukxHEsr3DRT8UyINw5Q4TCrOus+RcdaVe0thCUWUlG1SknQLKZXm/LLcwf1seLmcBAAW367uaeFZVhli6GMgwc98eK/zHKCJEF/xzuVuzFBh+QttOaLw9+w+DakMWaanqXwOuWY3h6pxIQ+BPDRt9Ocm
9/BioRdBp4tI88/xOhNjkSnyujPVg0TkzVAf47RyxnWAk/pOze181TfB7Fhpp1crgWyJe2TXf8oPLThrQeiFhypu24GvmzYmnLLJt23we3WQeuIzZsCY2esBl3xCYPEBlYthxuCnxLrZsz80sNIaI4JnywAH
JYSZgT3R3IgSX5YT5nRIREZFSrbz3gEzZI/fBEIKpxsziOlsGkOrgIx49zWasq412P1vMXjE/NAbnMG9YlgIEY3FkD+QGW0v8NL2eplqOKD8toeReRc2CEu4u599v8iLSauZzXIasfW3H3HzScF9n2/637kS
/YSvtOLPAeFQzDO/77W+lRCOzzcYkqPUeqr3q506d+Y+ejCaRlyRTW3W+1Q4LuqYbpDPK4YlAqJEJOJ2qsX1fOnPa0XdanAigpw== jouvensj01@gmail.com
PS C:\Users\Jouvens\Desktop\projet-git>
```

14) Ajout de la clé sur mon compte GitHub

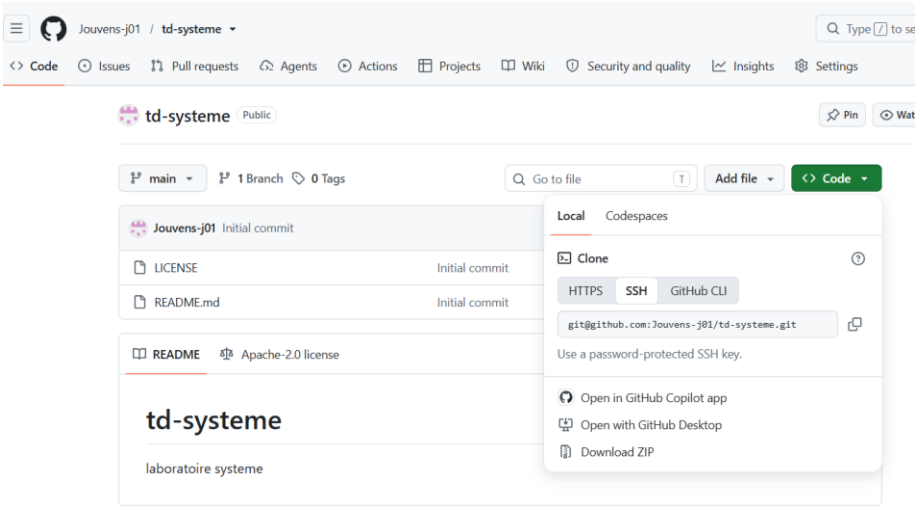




15) Création d'un dépôt sur GitHub



16) Clonage du dépôt avec la clé SSH




```
PS C:\Users\Jouvens\Desktop\projet-git> git clone git@github.com:Jouvens-j01/td-systeme.git
Cloning into 'td-systeme'...
The authenticity of host 'github.com (140.82.114.3)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), 4.77 KiB | 375.00 KiB/s, done.
PS C:\Users\Jouvens\Desktop\projet-git>
```

```
    Directory: C:\Users\Jouvens\Desktop\projet-git


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/15/2026   2:48 PM                td-systeme
-a----        10/15/2026   1:45 PM             26 README.md


PS C:\Users\Jouvens\Desktop\projet-git>
```

```
PS C:\Users\Jouvens\Desktop\projet-git> git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   Module.txt
        new file:   td-systeme

PS C:\Users\Jouvens\Desktop\projet-git>
```

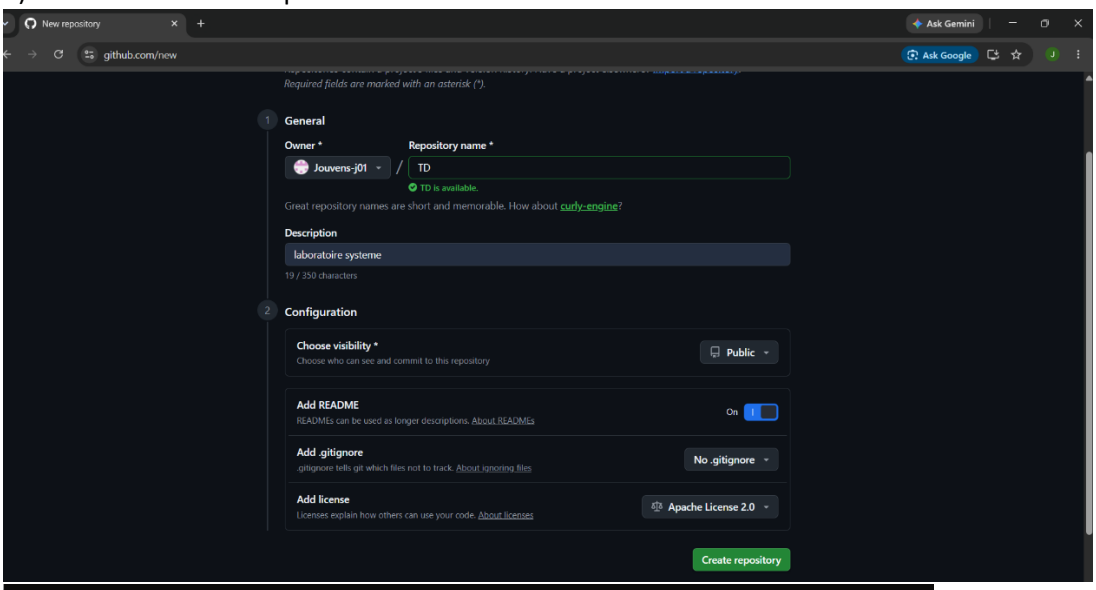
```
PS C:\Users\Jouvens\Desktop\projet-git> git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   Module.txt
        new file:   td-systeme

PS C:\Users\Jouvens\Desktop\projet-git> git commit -m "Ajout Module"
[master 4a0fdc2] Ajout Module
 2 files changed, 1 insertion(+)
 create mode 100644 Module.txt
 create mode 160000 td-systeme
PS C:\Users\Jouvens\Desktop\projet-git>
```

```
PS C:\Users\Jouvens\Desktop\projet-git> git push -u origin main --force
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 8 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (10/10), 918 bytes | 459.00 KiB/s, done.
Total 10 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/Jouvens-j01/td-systeme.git
 + b6aad83...97c7655 main -> main (forced update)
branch 'main' set up to track 'origin/main'.
PS C:\Users\Jouvens\Desktop\projet-git>
```

2eme partie du TD

1) Créer un nouveau dépôt sur GitHub




```
PS C:\Users\Jouvens\Desktop> git clone git@github.com:Jouvens-j01/TD.git
Cloning into 'TD'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), 4.76 KiB | 375.00 KiB/s, done.
PS C:\Users\Jouvens\Desktop> cd TD
PS C:\Users\Jouvens\Desktop\TD> mkdir systeme


    Directory: C:\Users\Jouvens\Desktop\TD


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/19/2026  12:59 PM                systeme
```

```
PS C:\Users\Jouvens\Desktop\TD> cd systeme
PS C:\Users\Jouvens\Desktop\TD\systeme> mkdir image


    Directory: C:\Users\Jouvens\Desktop\TD\systeme


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/19/2026   1:00 PM                image


PS C:\Users\Jouvens\Desktop\TD\systeme> mkdir projet


    Directory: C:\Users\Jouvens\Desktop\TD\systeme


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/19/2026   1:02 PM                projet
```

```
PS C:\Users\Jouvens\Desktop\TD\systeme> mkdir projet


    Directory: C:\Users\Jouvens\Desktop\TD\systeme


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/19/2026   1:02 PM                projet


PS C:\Users\Jouvens\Desktop\TD\systeme> mkdir presentation


    Directory: C:\Users\Jouvens\Desktop\TD\systeme


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/19/2026   1:03 PM                presentation


PS C:\Users\Jouvens\Desktop\TD\systeme> cd ..
PS C:\Users\Jouvens\Desktop\TD>
```

```
PS C:\Users\Jouvens\Desktop\TD> cd reseauI
PS C:\Users\Jouvens\Desktop\TD\reseauI> mkdir image


    Directory: C:\Users\Jouvens\Desktop\TD\reseauI


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/19/2026   1:05 PM                image


PS C:\Users\Jouvens\Desktop\TD\reseauI> mkdir projet


    Directory: C:\Users\Jouvens\Desktop\TD\reseauI


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/19/2026   1:07 PM                projet


PS C:\Users\Jouvens\Desktop\TD\reseauI> mkdir presentation


    Directory: C:\Users\Jouvens\Desktop\TD\reseauI
```

```
PS C:\Users\Jouvens\Desktop\TD\reseauI> mkdir presentation


    Directory: C:\Users\Jouvens\Desktop\TD\reseauI


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/19/2026   1:07 PM                presentation


PS C:\Users\Jouvens\Desktop\TD\reseauI> cd ..
PS C:\Users\Jouvens\Desktop\TD>
```

Conclusion

ce que j’ai appris en faisant ce travaux dirigé , comprends les concepts de base du Git et GitHub , la capacité a cloner un projet sur mon ordinateur puis héberger un projet sur GitHub avec PowerShell.

la list des commandes utilises pour faire le travail :

git --version : Permet de vérifier la version git installée sur son ordinateur.

git config –-global user.name et git config –-global user.email :

Permet de configurer le compte GitHub avec des données utilisateurs.

git config --list : Permet d’afficher la liste des configurations de Git.

ls : permet l’affichage de tous les répertoires.

cd : Permet l’ouverture d’un dossier.

mkdir : Permet de créer un dossier.

git init : Permet l’initialisation d’un nouveau dépôt git local dans le dossier actuel.

README.md : Fichier texte écrit en Markdown.

git log : Permet d’afficher l’historique des **commits** dans un dépôt git.

git status : Sert à afficher l’état actuel du dépôt Git.

git add . : Sert à ajouter des fichiers

git commit –m : Valide les changements ajoutés.

git branch –M main : Permet de renommer la branche actuelle en main, même si une branche main existe déjà.

git push –u origin main : Permet d’envoyer la branche main vers GitHub.

Le problème que j’ai rencontré en faisant la première partie de ce TD est déjà résolu. J’ai pu le push avec succès vers le dépôt GitHub.